

Objective

To implement and demonstrate Hierarchical Inheritance in Java, where multiple child classes (Dog and Cat) inherit common properties and methods from a single parent class (Animal).

Steps for Execution

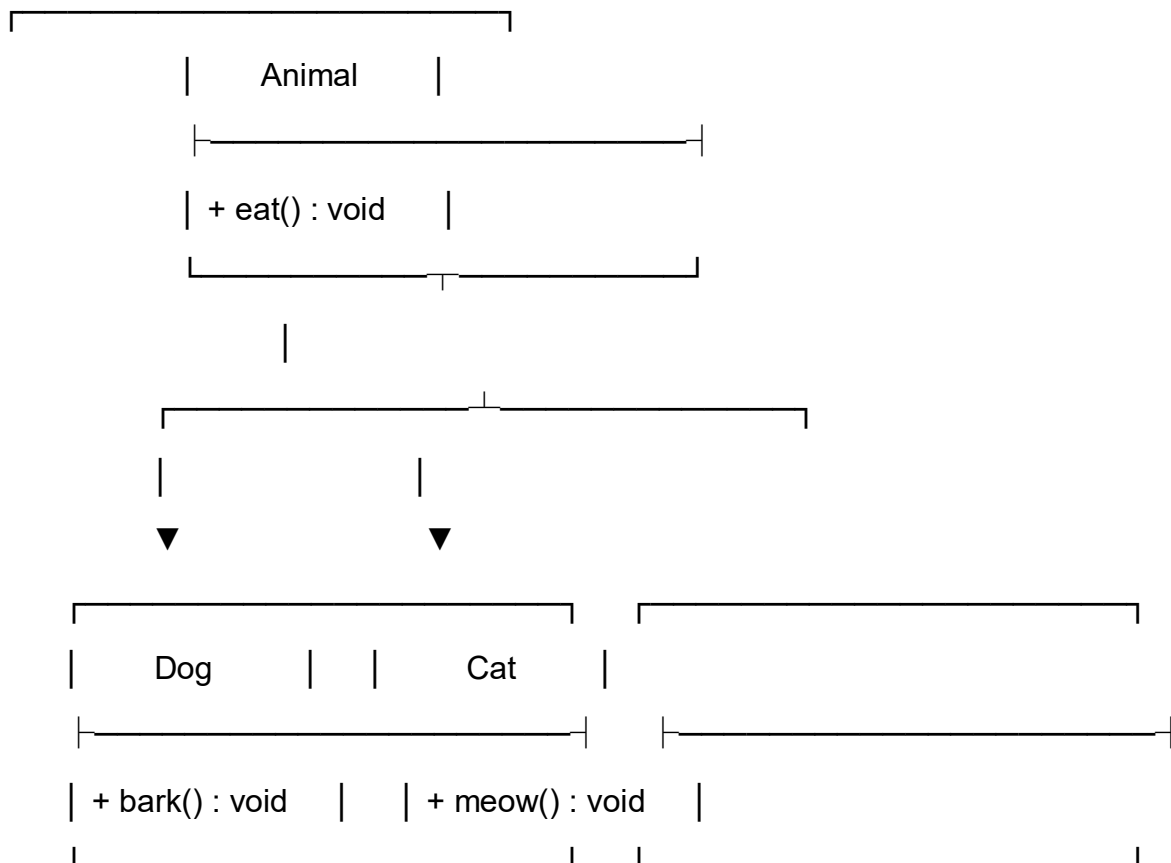
1. Open a Java IDE such as Eclipse, IntelliJ IDEA, or VS Code.
2. Create a package named javacore.
3. Create a Java file named HierarchicalInheritance.java.
4. Define the parent class Animal with the eat() method.
5. Define the child class Dog that extends Animal and contains the bark() method.
6. Define the child class Cat that extends Animal and contains the meow() method.
7. Create the main() method inside the HierarchicalInheritance class.
8. Create an object of Dog and call:
 - eat() inherited from Animal
 - bark() defined in Dog
9. Create an object of Cat and call:
 - eat() inherited from Animal
 - meow() defined in Cat
8. Compile and execute the program.
9. Observe the output in the console.

Algorithm

- Step 1: Start the program.
- Step 2: Define the superclass Animal with the method eat().
- Step 3: Define the subclass Dog by extending Animal.
- Step 4: Define the bark() method inside Dog.
- Step 5: Define the subclass Cat by extending Animal.
- Step 6: Define the meow() method inside Cat.
- Step 7: Create a Dog object.

- Step 8: Invoke eat() using the Dog object.
- Step 9: Invoke bark() using the Dog object.
- Step 10: Create a Cat object.
- Step 11: Invoke eat() using the Cat object.
- Step 12: Invoke meow() using the Cat object.
- Step 13: Display the results.
- Step 14: Stop the program.

UML Class Diagram



Inheritance Relationship

This is called Hierarchical Inheritance because two or more subclasses inherit from one superclass.

Animal

/ \

/ \
Dog Cat

Code:

```
package javacore;

class Animal {
    void eat() {
        System.out.println("Animal eats");
    }
}

class Dog extends Animal {
    void bark() {
        System.out.println("Dog barks");
    }
}

class Cat extends Animal {
    void meow() {
        System.out.println("Cat meows");
    }
}

public class HierarchicalInheritance {
    public static void main(String[] args) {
        Dog d = new Dog();
        d.eat();
        d.bark();
        Cat c = new Cat();
        c.eat();
        c.meow();
    }
}
```

```
}
```

Output Screenshot(sample Run):

Animal eats

Dog barks

Animal eats

Cat meows

Result:

The program successfully demonstrates Hierarchical Inheritance in Java. The Dog and Cat classes inherit the eat() method from the common superclass Animal, while each child class has its own specific method (bark() or meow()).